

Further Statistics 1

Notes · Edexcel A Level Further Mathematics — Further Statistics 1 · Chapter 3: Geometric and Negative Binomial Distributions

Chapter 3: Geometric and Negative Binomial Distributions

Why this chapter matters

So far every distribution has counted *how many* successes occur. This chapter flips the question around: given a sequence of independent trials, how many trials do we need before we see the first success (Geometric), or before we see the r -th success (Negative Binomial)? These “waiting time” distributions come up naturally in quality control, games of chance, and reliability testing, and they give you two more named formulae to add to your toolkit alongside Binomial and Poisson.

Concept: conditions for a Geometric model

$X \sim \text{Geo}(p)$ models the number of independent Bernoulli trials, each with constant probability of success p , up to and including the **first** success. The conditions are:

1. Each trial has exactly two outcomes (success/failure).
2. The probability of success p is constant from trial to trial.
3. Trials are independent.
4. X counts the trial number on which the first success occurs (so $X = 1, 2, 3, \dots$).

Key formula: Geometric probability, mean and variance

KEY FORMULA	
Quantity	Formula
$P(X = x)$	$p(1 - p)^{x-1}$, for $x = 1, 2, 3, \dots$
$E(X)$	$1/p$
$\text{Var}(X)$	$(1 - p)/p^2$
$P(X > x)$	$(1 - p)^x$

The result $P(X > x) = (1 - p)^x$ is extremely useful and worth memorising separately — it says “the probability of needing more than x trials is just the probability of x consecutive failures,” which is obvious once you see it, but saves a lot of summing in exams.

Taking the complement the other way round gives $P(X \leq x) = 1 - (1 - p)^x$, which is what you need for “within x trials” or “at most x trials” questions — don’t try to sum $p(1-p)^0 + p(1-p)^1 + \dots + p(1-p)^{x-1}$ term by term when this shortcut does it in one line.

A convention to watch for: this course defines X as the trial number of the first success, so X starts at 1 ($P(X = 0)$ is not defined). Some other textbooks and software (including most calculators’ built-in

Geometric functions) instead define a variable $Y = X - 1$ as the number of *failures before* the first success, starting at $Y = 0$ — if you ever check an answer with a calculator's Geo/NegBinom function and the mean is out by exactly 1 (or r), this is almost always why.

Worked example 1: basic Geometric probability

EXAMPLE 3.1

Question. A biased coin has $P(\text{heads}) = 0.3$. The coin is tossed repeatedly until a head is obtained. Let X be the number of tosses required. Find (a) $P(X = 4)$, (b) $P(X > 5)$.

Step 1. State the model: $X \sim \text{Geo}(0.3)$. **(B1)**

Step 2. (a) $P(X = 4) = (0.3)(0.7)^3 = 0.3 \times 0.343 = 0.1029$. **(M1A1)**

Step 3. (b) $P(X > 5) = (0.7)^5 = 0.16807$. **(M1A1)**

Answer: (a) 0.103 (3 s.f.), (b) 0.168 (3 s.f.)

EXAM TIP

It is very common to confuse $\text{Geo}(p)$ with $B(n, p)$ because both involve repeated trials with a fixed success probability. The structural difference: Binomial fixes the *number of trials* and counts successes; Geometric fixes the target as “first success” and the number of trials is the random variable. A quick diagnostic question (“is n fixed here, or is it what we’re being asked to find?”) resolves most confusion.

Concept: the memoryless property

The Geometric distribution has the memoryless property: $P(X > s + t \mid X > s) = P(X > t)$. In words, if you have already had s failures, the probability of needing at least t more trials is exactly the same as it was at the start — previous failures give no information about how much longer you’ll wait. This is a popular “explain why” or “show that” exam question.

Worked example 2: memoryless property

EXAMPLE 3.2

Question. Using the coin from Example 1 ($P(\text{heads}) = 0.3$), given that the first 3 tosses were all tails, find the probability that at least 2 more tosses are needed to get a head.

Step 1. “At least 2 more tosses are needed” after the first 3 tails means $X > 4$ (the first head still hasn’t appeared by toss 4), given that we already know $X > 3$. By the memoryless property, $P(X > 4 \mid X > 3) = P(X > 1)$, since the gap between 4 and 3 is 1. **(M1)**

Step 2. $P(X > 1) = (0.7)^1 = 0.7$. **(A1)**

Answer: 0.7

EXAM TIP

Rather than deriving the memoryless property algebraically every time ($P(X > s+t \mid X > s) = P(X > s+t)/P(X > s) = (1-p)^{(s+t)}/(1-p)^s = (1-p)^t$), which is good to see once, build the intuition with a physical analogy: a coin has no memory of how many times it has already come up tails, so “waiting for the next success” always resets to looking exactly like a fresh Geometric

distribution.

Concept: the Negative Binomial distribution

$X \sim \text{NB}(r, p)$ models the number of trials needed to obtain the r -th success, under the same conditions as Geometric (independent trials, constant success probability p). Geometric is simply the special case $r = 1$.

Key formula: Negative Binomial probability, mean and variance

For $X \sim \text{NB}(r, p)$, the number of trials up to and including the r -th success:

KEY FORMULA

Quantity	Formula
$P(X = x)$	$(x-1 \text{ choose } r-1) \cdot p^r \cdot (1-p)^{x-r}$, for $x = r, r+1, r+2, \dots$
$E(X)$	r/p
$\text{Var}(X)$	$r(1-p)/p^2$

The combinatorial factor $(x-1 \text{ choose } r-1)$ accounts for the fact that the r -th success must occur on trial x , while the *other* $r-1$ successes can occur anywhere among the first $x-1$ trials.

It's worth seeing where $E(X) = r/p$ and $\text{Var}(X) = r(1-p)/p^2$ come from, rather than treating them as new formulae to memorise: the number of trials to the r -th success is just the number of trials to the 1st success, plus the number of *additional* trials to the 2nd, plus ... plus the number of additional trials to the r -th — that is, X is the sum of r independent $\text{Geo}(p)$ variables (each “waiting time” between consecutive successes has exactly the same Geometric distribution, by the memoryless property). Summing r independent copies multiplies both the mean and the variance of a single $\text{Geo}(p)$ by r , giving $r \cdot (1/p) = r/p$ and $r \cdot (1-p)/p^2$ immediately.

Worked example 3: Negative Binomial probability

EXAMPLE 3.3

Question. A basketball player has a free-throw success probability of 0.6 on each independent attempt. Find the probability that her 3rd successful free throw occurs on her 5th attempt.

Step 1. Identify $r = 3$ (the 3rd success), $p = 0.6$, and we want $P(X = 5)$ where $X \sim \text{NB}(3, 0.6)$. **(B1)**

Step 2. Use the formula: $P(X = 5) = (5-1 \text{ choose } 3-1) \cdot (0.6)^3 \cdot (0.4)^{5-3} = (4 \text{ choose } 2) \cdot (0.6)^3 \cdot (0.4)^2$. **(M1)**

Step 3. Compute: $(4 \text{ choose } 2) = 6$, $(0.6)^3 = 0.216$, $(0.4)^2 = 0.16$, so $P(X = 5) = 6 \times 0.216 \times 0.16 = 0.20736$. **(M1A1)**

Answer: 0.207 (3 s.f.)

EXAM TIP

The most reliable way to check a Negative Binomial calculation by hand is to reason about it directly: “the 5th attempt **MUST** be a success (probability p), and among the first 4 attempts exactly 2 must be successes” — this reframes the formula as (choose which 2 of the first 4 are successes) $\times$ (probability of that particular arrangement) $\times$ (probability the 5th is also a success), which is exactly $p^3(1-p)^2 \times (4 \text{ choose } 2)$. Working through this derivation yourself demystifies the formula far more than presenting it as something to memorise.

Chapter summary

Idea	Key point
Geometric distribution	Number of trials to the first success; $P(X=x) = p(1-p)^{x-1}$
Geometric mean/variance	$E(X) = 1/p$, $\text{Var}(X) = (1-p)/p^2$
$P(X > x)$ shortcut	$P(X > x) = (1 - p)^x$ (x consecutive failures); $P(X \leq x) = 1 - (1 - p)^x$
Memoryless property	$P(X > s+t \mid X > s) = P(X > t)$
Negative Binomial distribution	Number of trials to the r -th success; generalises Geometric ($r = 1$)
Negative Binomial mean/variance	$E(X) = r/p$, $\text{Var}(X) = r(1-p)/p^2$

Common exam mistakes

- Confusing Geometric (counts trials to first success, $X = 1, 2, 3, \dots$) with Binomial (counts successes in a fixed number of trials).
- Using x instead of $x - 1$ in the exponent of $(1 - p)$ in the Geometric formula.
- Forgetting the combinatorial factor ($x-1 \text{ choose } r-1$) in the Negative Binomial formula, or using ($x \text{ choose } r$) instead.
- Misapplying the memoryless property to distributions other than Geometric (it does NOT hold for Negative Binomial with $r > 1$, nor for Binomial or Poisson).
- Forgetting that Negative Binomial requires the **LAST** trial (trial x) to be a success — allowing the r -th success to occur anywhere in the first x trials.

You should now be able to

- State the conditions for a Geometric model and calculate $P(X = x)$, $P(X > x)$ and $P(X \leq x)$.
- Explain and apply the memoryless property of the Geometric distribution.
- State the conditions for a Negative Binomial model and recognise it as a generalisation of Geometric.
- Calculate Negative Binomial probabilities using the formula involving ($x-1 \text{ choose } r-1$).

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- Quote and use the mean and variance formulae for both distributions without needing to re-derive them.